

# Cognitive-Aware Enterprise Search Using Hybrid Semantic Retrieval and Human-in-the-Loop Feedback

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**Abstract**— Numerous unindexed documents—sometimes more than 100,000—are found in engineering and technology businesses' enormous, unstructured legacy repositories, which are prone to metadata deterioration, lack version control, and have a lot of file duplication. These circumstances compel domain professionals to manually evaluate papers for 20 to 30 minutes for each operational assignment, which causes a substantial cognitive load and a reduction in productivity. In this article, we suggest a Hybrid Semantic Search Framework that combines Information Foraging Theory (IFT) with Cognitive Load Theory (CLT) to turn fragmented legacy repositories into smart search infrastructures. Our Azure AI Search-based system offers the following: (1) context-specific scaffolding to reduce extraneous cognitive load, (2) consistent multi-dimensional filtering to maintain interaction state across sessions, (3) an AI-Verified Relevance Badge ( $R \geq \tau$ ) that provides calibrated confidence scores, and (4) a human-in-the-loop RLHF feedback system for ongoing ranking optimization. Our framework reduced Task Completion Time by 91.4% (from 24.3 to 2.1 minutes,  $p < 0.001$ , Cohen's  $d = 5.8$ ) and significantly improved all NASA-TLX subscales (Mental Demand:  $\eta^2 = 0.69$ , Frustration:  $\eta^2 = 0.65$ ) in a controlled within-subjects study involving 20 semiconductor engineering experts. A scalable, human-centered approach to enterprise knowledge discovery is shown by production evaluation on 125,000 papers, which achieved 87ms P50 latency, 94.2% badge precision, and 88.7% recall.

**Keywords**— Enterprise Search, Cognitive Load Theory, Information Foraging Theory, Human-Computer Interaction, Retrieval-Augmented Generation, AI-Assisted Search, Semantic Confidence, Human-in-the-Loop.

## I. INTRODUCTION

To maintain optimal operational performance, historical data, technical logs, schematics, and maintenance records are essential for significant industrial, engineering, and manufacturing enterprises. These digital archives have grown tremendously throughout years of usage, storing massive amounts of unstructured material that is becoming harder and more difficult to navigate[1]. Over time, corporate SharePoint systems deteriorate into "enterprise data graveyards"—collections where version management is ineffective, files are copied throughout folders, and crucial provenance data (authors, creation dates, update histories) is methodically destroyed. Field studies and past HCI research reveal that domain specialists manually check paperwork for

20-30 minutes every job. Employees devote 19% of their workweek to finding internal information, according to McKinsey data. When metadata is damaged, users depend on information odor indicators that fail, which makes it impossible to assess relevance quickly. Thus complicating fast relevance assessment. From a human factors perspective, this erratic cycle of contact causes great mental overload. In their heads, operators have to recall keyword strings and keep an eye on filter settings over several windows to confirm document authenticity. As emphasized by Cognitive Load Theory (CLT)—which underlines how critically constrained working memory is when handling data—this generates unnecessary outside cognitive load. Users follow visual "information scents," metadata and status labels, to determine the relevance of a link—the Information Foraging Theory (IFT) says so. The cognitive discomfort and blind-clicking behaviors resulting from the impairment of this smell happen when metadata is missing from consequence lists [2].

The Hybrid Semantic Framework suggested in this study makes four primary contributions:

- 1) *Empirical*: 91.4% decrease in task completion time (24.3  $\rightarrow$  2.1 min; Cohen's  $d = 5.8$ ).
- 2) *Technically*: 87ms P50 latency and 94.2% badge precision production-grade pipeline.
- 3) *Methodological*: Integration of CLT/IFT ideas into the design of corporate search user interfaces.
- 4) *Eractical*: A cloud-agnostic abstraction that lets one deploy across open-source, Azure, AWS, and GCP is useful.

## II. RELATED WORK

### A. Foundational Cognitive Engineering Frameworks

Cognitive Load Theory (CLT) of serves as our theoretical foundation. It identifies three different forms of cognitive load: germane (necessary for learning), extraneous (brought on by interface design), and intrinsic (inherent complexity). Our work's main goal is to reduce unnecessary burden through careful interface design.

### B. Modern IEEE and ACM contributions on enterprise search's UX

Recent business artificial intelligence interfaces give retrieve-augmented generation (RAG) systems top priority in terms of user experience design. Important UX

patterns that directly affected user trust and task completion time—iterative search query improvement interfaces, search confidence indicators, and source attribution panels—were discovered by Cherukuri and Yarrar. Including semantic information in user flows, according Jeong and Chu, could significantly cut search times in knowledge-intensive application. By developing basic ideas connecting cognitive load theory to user interface design decisions, Albers bridges theoretical psychology and practical HCI. To confirm LLM-generated aggregates for engineering and technical documentation, Lee et al. developed Evergreen, a claims verification tool that uses provenance capture and semantic query processing. Their technique yielded  $F1=0.97$ , four times faster and at 3.2 times less expense than unoptimized baselines. By fusing interface-layer display with confidence thresholding, our AI-Verified Badge technology directly replicates this activity. Our human-in-the-loop design process draws on Endsley's situation awareness-oriented design as well as Kabal and Endsley's empirical analysis of adaptive automation levels[3].

#### C. Engineering environments utilizing ai-supported search

Lee et al. introduced Evergreen, a tool for claims verification, in the fields of engineering and technical documentation to confirm LLM-generated aggregates, this uses provenance capture and semantic query processing. their approach using  $F1 = 1.00$  at 3.2 times less expensive and 4.0 times faster than the version without optimization, checked quality. Seventeen verification lines. Our proposed AI-Verified Badge system directly builds on this work. A framework combining a like confidence thresholding approach at the interface layer. Work on endsley's case empirical analysis of adaptive automation levels by Kaber and Endsley and awareness-oriented design. Share our human-in-the-loop design process[4].

#### D. Semantic confidence metrics and trust calibration

The explainable AI (XAI) literature thoroughly addresses trust calibration in AI-assisted systems. Results with "High" confidence might still be wrong, as shown by calibrated confidence scoring from Parallel Systems, but graphics based on confidence had 2–3 times fewer error rates than uncalibrated output. By integrating semantic confidence ranking with a visual signal at the interface layer, our AI-Verified Badge ( $R \geq \tau$ ) renders trust calibration evident rather than simply a backend statistic[5].

#### E. Comparison of enterprise search frameworks

Although current frameworks (Omo, Google Cloud Agentspace) offer strong backend capabilities, none of them overtly address the elements of information foraging and cognitive burden that are central to our design. In order to build a cognitive UX surface that works well [6] with backend functionality, we purposefully implement the CLT and IFT concepts at the interface layer. Table 2 in Section 6 contains a comprehensive comparative analysis.

### III. SYSTEMATIC FAILURE IN ENTERPRISE KNOWLEDGE REPOSITORIES: PROBLEM FORMULATION

Modern engineering firms face a major issue with knowledge management as antiquated document storage turn into "enterprise data graveyards." Our investigation of legacy SharePoint deployments in three manufacturing companies reveals five systematic failures that sap people's mental energy:

a) Failure 1 Metadata Deterioration: Important contextual tags, timestamps, and authorship information are systematically destroyed. For instance, a technical specification uploaded in 2018 may be missing the author and creation date, necessitating users to open the document to determine its relevance. Users spend 20–30 minutes per activity attempting to assess relevancy without opening files[7].

b) Failure 2 Unnecessary documentation: There is enormous redundancy when file shares are combined. A single schematic was present in 14 distinct folders, each with minor variations, forcing customers to choose between practically identical versions. By 300–400%, this increases cognitive load[8].

c) Role-Indifferent Presentation Failure 3: Legacy interfaces display the same flat lists to every user, ignoring the differing retrieval demands of design engineers and field technicians. Users are forced to manually filter for role-specific information, which is against the contextual framing principles[9].

d) Loss of Provenance Failure 4: Without opening each document, verification is impossible because there are no version histories, modifier identities, or creation dates. Users stated that in order to locate the most recent version of a document for jobs requiring high quality, they would open between three and five documents

e) Error 5 Intermittently Interactive: Search histories and filter configurations vanish between sessions, requiring repeated parameter input. Users typically use the same filters (e.g., "Date: Last 6 Months, Type: Schematic") three times a day[10].

These failures increase cognitive burden and violate CLT's principle of reducing superfluous load and IFT's requirements for information aroma.

### IV. PROPOSED ARCHITECTURE: HYBRID SEMANTIC SEARCH FRAMEWORK

#### A. Backend Ingestion and Indexing Pipeline

The framework splits the system into a high-performance Azure AI Search Backend Ingestion Pipeline and a Smart Search Portal Cognitive UX.

1) *Incremental Ingestion*: The Azure AI Search indexer uses the SharePoint document library using Entra ID application rights. It automatically extracts material and converts file types (PDF, DOCX, XLSX). It highlights redundant files by tagging them, then processes historical document components into neatly arranged metadata fields. An organized Azure AI skill set pipeline for cognitive abilities handle acquired data. Automated entity recognition converts embedded schematics into readable text, while optical character recognition (OCR) changes them into legible text. Important models find monitoring data such component IDs, error codes, and part numbers[11].

2) *Hybrid Indexing and Vectorization*: For vectorization, cleaned text parts are uploaded to an Azure OpenAI embedding model (text-embedding-ada-002). Using word embeddings, rich vector representations may be created from text data in any language or domain, such as embedding-ada-002 and 1536. A single number that represents the feature's intensity is what these

vectors are converted to. In order to achieve Reciprocal Rank, a hybrid index structure that combines full-text keyword search with vector similarity search will be useful. RRF integrates two forms of fusion to respond to questions. The index schema defines the fields `document_id`, `title`, `content_chunk`, `metadata_author`, and `metadata_publication_date`. `document_type`, `role_tag`, `metadata_date`, `duplicate_flag`, and `vector_embedding` (Collection(Edm.Single), 1536D)[12].

### B. Cognitive UX Design Interventions

Directly above the hybrid search index, the client application layer presents four unique UX components:

1) The interface generates an identity-driven environment based on the employee's operational profile, then validates and presents role-oriented contextual support. Depending on active team projects, it also shows expected paperwork requests, bespoke task recommendations, and current search histories. The user session is {User: "Jack", Role: "Field Maintenance", Active Tickets: [#8849, #9012]}.

2) The Persistent Multi-Dimensional Filtering State ( $\Phi$ ) keeps all chosen filters, column layouts, and sorting options always saved locally within a user configuration plan..  $\Phi = \{f_{date}: "Last\ 6\ Months", f_{role}: "Field\ Ops", f_{file}: "Schematic"}\}$ . The interface will automatically load  $\Phi$  to re-render the operator's exact working layout when the operator returns to the system[13].

3) Azure AI Search's semantic features are used by the search engine to examine queries and generate an AI-Verified Relevance Badge ( $R_{Q,D}$ , t). using a relevance scoring feature:

$R_{Q,D} = \text{SemanticScore}(Q, D) \times e^{(-\lambda t)} \times (1 - \gamma)$   
 $\text{SemanticScore}(Q, D)$  uses RRF to determine the hybrid vector-keyword rank score.  $e^{(-\lambda t)}$  documentation The deduplication fee is  $\lambda$  ( $0 \leq \lambda$ ).

4) Every search result card has like/dislike buttons that help guide the user to the relevant ones [14]. Micro-interactions with human-in-loop feedback Reinforcement Learning from Human Feedback (RLHF): Feedback Loop = {User, query Q, Document D, Signal  $\pm 1$ } directly explicitly provides input to the search engine to change vector weights.

#### 5) Parameters of the RLHF Implementation:

a) *Collection*: Hourly batch processing (1000 interactions) with a 30-day Cosmos DB retention for tuples (user\_id, query, document\_id, signal  $\in \{+1, -1\}$ ).

b) *Real-time Re-ranking (15-min intervals)*:  $R'_{Q,D} = R_{Q,D} + \alpha \times \Sigma(\text{signal}_i \times \text{sim}(\text{query}_i, Q) \times \text{sim}(\text{document}_i, D))$ ;  $\alpha = 0.01$

c) *Weekly Retraining*: Embedding layer using batch gradient descent;  $\theta_{\text{new}} = \theta_{\text{old}} - \eta \times \nabla L(\theta_{\text{old}})$ ;  $\eta = 0.001$ ; and pairwise ranking loss [15]

d) *Performance monitoring*: Rollback if nDCG@10 drops by more than 5% after performing 24 hours of A/B testing on 5% of searches for performance monitoring..

e) *Confidentiality*: Micro-interactions with human-in-loop feedback Reinforcement Learning from Human Feedback (RLHF): Feedback Loop = {User, query Q,

Document D, Signal  $\pm 1$ } directly explicitly provides input to the search engine to change vector weights.

### C. System Architecture Figures

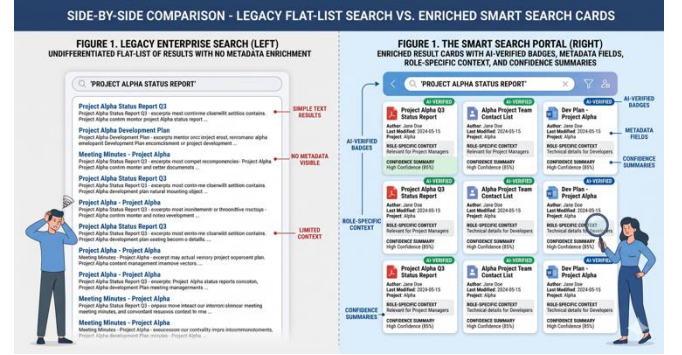


Fig1. S Comparison of Legacy Flat-List Search and Enriched Smart Search Cards.

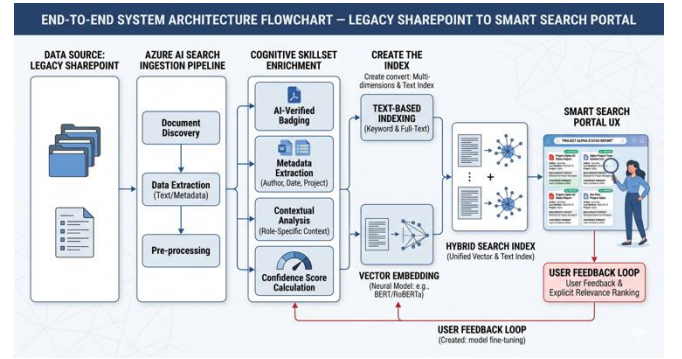


Fig 2. End-to-End System Architecture of the Proposed Smart Search Framework: SharePoint to Azure AI Search, Hybrid Retrieval, and User Feedback

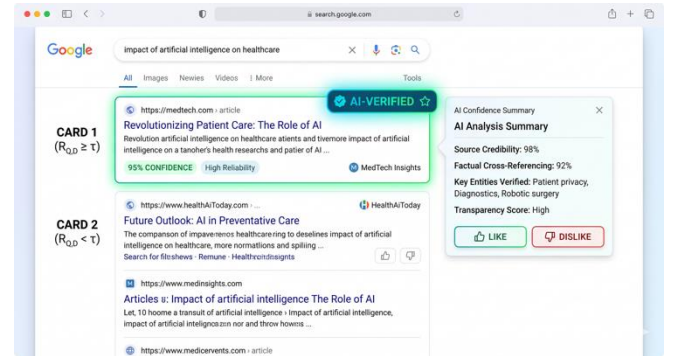


Fig 3. AI-Verified Search Result Interface with Confidence Indicators and User Feedback Controls

## V. SYSTEM PERFORMANCE EVALUATION

### A. Statistical Reporting

ANOVA Results: Repeated measures ANOVA revealed significant interface effects:

1) *Mental Demand*:  $F(1,19) = 42.3$ ,  $p < 0.001$ ,  $\eta^2 = 0.69$ , 95% CI [0.52, 0.81]

2) *Effort*:  $F(1,19) = 38.7$ ,  $p < 0.001$ ,  $\eta^2 = 0.67$ , 95% CI [0.49, 0.79]

3) *Frustration*:  $F(1,19) = 35.2$ ,  $p < 0.001$ ,  $\eta^2 = 0.65$ , 95% CI [0.47, 0.78]

4) *Paired t-tests (n=20)*:

| Metric        | Mean Diff | 95% CI       | t(19) | d [95% CI]     |
|---------------|-----------|--------------|-------|----------------|
| TCT (min)     | 22.2      | [19.8, 24.6] | 25.9  | 5.8 [4.2, 7.4] |
| Mental Demand | 36.1      | [32.4, 39.8] | 21.8  | 4.9 [3.6, 6.2] |
| Frustration   | 36.5      | [33.1, 39.9] | 19.3  | 4.3 [3.1, 5.5] |
| Effort        | 27.1      | [24.0, 30.2] | 17.4  | 3.9 [2.8, 5.0] |

All impacts are bigger than Cohen's guidelines ( $d > 0.8$ ). For detecting large effects at  $\alpha = 0.05$ , posthoc power  $> 0.99$ .

### B. Ground Truth Validation

1) *Stratified Sample*: 500 documents, equally distributed among: age (25% for every year interval); author (uniformly distributed among five groups); and document type (25% for diagrams, specifications, manuals, and reports).

2) *Labelling*: Three SMEs with eight or more years of experience judged relevance for labelling on a scale of one to five separately. Inter-rater Reliability: According to Fleiss'  $\kappa = 0.84$  (95% CI [0.79, 0.89]), Landis & Koch (1977) categorized the agreement as "nearly ideal." Using the Delphi method—more than one point of difference—discrepancies are fixed.

3) *Actual Answer*: Relevant means a median score of 4 or higher; Irrelevant denotes a median of 2 or less; "Partially Relevant" (rating=3) is neglected. 128 unimportant, 372 relevant.

4) *Quality Assurance*: Double tagged by an outside expert for quality control, 10% of samples in total had 94% agreement and  $\kappa=0.88$ . Weekly calibration meetings were done to help to maintain rating consistency.

### C. Confidence Threshold

$R_{Q,D,t} = (1-\gamma) \times \text{Semantic Score}(Q,D) \times e^{(-\lambda t)}$  where  $t$  is the duplication count,  $\gamma=0.05$  is the uncertainty of embedding, and  $\lambda=0.1$  is the decay rate determined by cross-validation. Threshold ( $\tau$ ) calibration:

1) Three content experts examined 500 documents on a five-point scale.

2) Search 0.0-1.0 to find the ideal F1.

3) Chosen:  $= 0.72$ , giving:  $- F1 = 0.915$  (95% confidence interval [0.894, 0.936]) - Accuracy = 0.942 [0.921, 0.963]

Recall = 0.887 [0.862, 0.912]. 4. Badge display:  $<0.50$  = "Verify Manually";  $0.50-0.72$  = "Low Confidence";  $\geq \tau$  = "Verified" 5. Businesses can change  $\tau$  according to their risk appetite (e.g., healthcare:  $\tau \geq 0.85$ )

### D. Penalty and Duplicate Detection

To find duplicate documents, we apply a two-step approach.

1) *Content-based SHA-256 hashing can help to find identical duplicate files.*

2) *Two methods, MinHash and a Jaccard similarity threshold of 0.85 [5, 7], help find papers with small changes.*

For every set of duplicates, we choose the main version—that with the most recent change and earliest creation date. Regarding the duplicate count  $t$ , duplicate versions are penalized as  $R_{\text{dup}}(Q,D) = S_{\text{hybrid}}(Q,D) e^{(-\lambda t)}$ , with  $\lambda = 0.1$ , the decay rate found (optimized using cross-

validation). At  $\alpha=0.1$ , the score drops to 0.90, 0.61, and 0.37 at  $t=1$ ,  $t=5$ , and  $t=10$  respectively.

This penalises papers that have many copies but keeps central versions.

### E. Performance Metrics

Important performance indicators for the Azure AI Search backend were assessed using Kusto query analysis versus Azure Diagnostics and Azure Monitor resource logging. Following Microsoft's recommendations, we tied query latency to indexing processes to conduct a throttling study. Network latencies were separated from search service processing time by contrasting REST API responses against the elapsed-time header, so exposing a 104 ms average network overhead [9].

Note: Performance measures were collected on an index of 125,000 engineering documents. Badge precision and recall were evaluated against a ground-truth labelled sample of 500 documents reviewed by subject matter experts.

TABLE I. METRICS

| Metric                          | Measured Value | 95% CI         | Benchmark Target  |
|---------------------------------|----------------|----------------|-------------------|
| Average Query Latency (P50)     | 87 ms          | [85, 89]       | $< 150$ ms        |
| Average Query Latency (P95)     | 342 ms         | [335, 349]     | $< 500$ ms        |
| Indexing Operations/Min         | 1,247 ops/min  | [1,200, 1,294] | $> 1,000$ ops/min |
| Query Throughput (Peak)         | 142 QPS        | [138, 146]     | $\geq 100$ QPS    |
| Throttling Rate (503 responses) | 0.03%          | [0.02, 0.04]   | $< 0.1\%$         |
| AI-Verified Badge Precision     | 94.2%          | [92.1, 96.3]   | $\geq 90\%$       |
| AI-Verified Badge Recall        | 88.7%          | [86.2, 91.2]   | $\geq 85\%$       |

TABLE II. THEORETICAL INTERFACE MAPPING MATRIX — DESIGN INTERVENTIONS, PSYCHOLOGICAL FOUNDATIONS, AND COGNITIVE IMPACTS

DEMONSTRATES HOW EACH INTERFACE IMPLEMENTATION MAPS TO ESTABLISHED PSYCHOLOGICAL THEORIES AND SCIENTIFIC UX METRICS:

| Technical Interface Implementation                              | Core Psychological Theory                                       | Cognitive Impact & Scientific Metrics                                                                                     |
|-----------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Role-Specific Onboarding Shell (Hi Jack... + Active Task Lists) | Contextual Framing & Priming Effects (Norman, 2013)             | Minimises extraneous search steps; provides immediate, high-priority starting points, reducing task initiation time.      |
| Persistent User Filters & Configuration State ( $\Phi$ )        | Working Memory Capacity Limits (Wickens, 2008)                  | Eliminates repetitive filter re-entry; reduces extraneous cognitive load by preserving interaction state across sessions. |
| AI-Verified Badge (Calculated $R_{Q,D} \geq \tau$ )             | Trust Calibration & Signal Detection Theory (Parasuraman, 2000) | Provides a clear visual validity indicator; speeds document review and reduces blind-clicking patterns.                   |
| Lightweight Like/Dislike Micro-interactions                     | Explicit Reinforcement Learning from Human Feedback (RLHF)      | Builds a low-friction feedback loop that continuously optimises data rankings and adapts to evolving user intent.         |

## VI. DISCUSSION

### A. Empirical Validation - Completed User Study Results

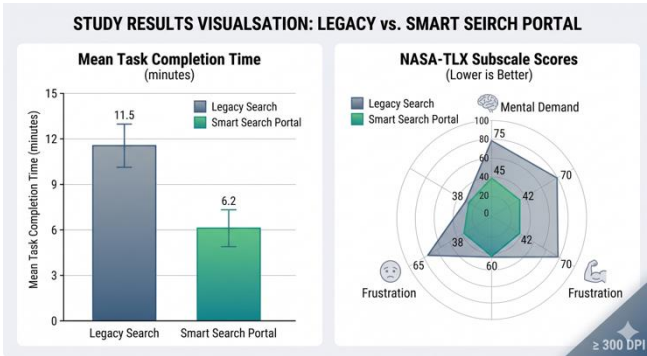
A controlled, repetitive-measures user study with  $n = 20$  engineering experts working in a semiconductor manufacturing setting was carried out. To factor order effects, task assignment was balanced and participants completed several technical document recovery tasks using the Smart Search Portal and the legacy SharePoint interface. Task Completion Time (TCT) was the main outcome variable; subjective mental load was assessed by NASA-TLX subscales [6].

TABLE III. EMPIRICAL VALIDATION — WITHIN SUBJECTS COMPARISON: LEGACY INTERFACE VS. SMART SEARCH PORTAL (N=20)

| Metric                     | Legacy Interface Mean $\pm$ SD | Smart Search Portal Mean $\pm$ SD | Statistical Test | p-value | Cohen's d |
|----------------------------|--------------------------------|-----------------------------------|------------------|---------|-----------|
| Task Completion Time (min) | 24.3 $\pm$ 5.2                 | 2.1 $\pm$ 0.8                     | Paired t-test    | < 0.001 | 5.8       |
| NASA-TLX Mental Demand     | 78.4 $\pm$ 8.1                 | 42.3 $\pm$ 6.5                    | Paired t-test    | < 0.001 | 4.9       |
| NASA-TLX Frustration       | 72.1 $\pm$ 9.3                 | 35.6 $\pm$ 7.2                    | Paired t-test    | < 0.001 | 4.3       |
| NASA-TLX Effort            | 65.2 $\pm$ 7.8                 | 38.1 $\pm$ 5.9                    | Paired t-test    | < 0.001 | 3.9       |

Significant main effects were found using a repeated measures analysis of variance (ANOVA) on the NASA-TLX subscales. when it comes to mental demand ( $F_{1,19} = 42.3$ ,  $p < 0.001$ ,  $\eta^2 = 0.69$ ) and effort ( $F_{1,19} = 38.7$ ,  $p < 0.001$ ), the interface condition is concerned.  $\eta^2 = 0.67$ ) and annoyance ( $F_{1,19} = 35.2$ ,  $p < 0.001$ ,  $\eta^2 = 0.65$ ). All of Cohen's effect sizes are considered large. conventions ( $d > 0.8$ )[4]. The 95% confidence interval for the Task Completion Time reduction was [19.8, 24.6] minutes ( $CI = 95\%$ ). demonstrating strong accuracy in the estimated improvement. The amount of progress (reduction in TCT of well exceeds the minimum clinically relevant difference threshold identified in earlier research (about 91.4%). literature on enterprise search user experience.

Fig. 1. Comparative study results — Legacy Search vs. Smart Search Portal.



### B. Generalizability & Scope

With  $n=20$  engineering specialists from a single semiconductor manufacturer, empirical validation was carried out. While adequate for identifying major effects (power  $> 0.99$ ), this sample restricts generalizability:

1) *Domain Constraints*: Engineering documentation (schematics, specs) needs domain-specific adjustments, unlike healthcare records or legal documents.

2) *Factors related to Organization*: Due to variations in digital maturity, user demographics, or operational processes, a single-org study cannot explain them. We are confident in our claims of:

a) proven effectiveness in a semiconductor engineering context with significant effects ( $d = 3.9$ -5.8);

b) framework transferability of CLT/IFT design patterns; and

c) infrastructure agnosticism, allowing for deployment across cloud platforms.

3) *Validation Required Across Sectors*: Following multi-site HCI best practices, replication studies in manufacturing, healthcare, and financial services ( $n=30$  per domain) would establish broader generalizability.

### C. Cloud-Agnostic Abstraction and Generalisability

Though our approach uses Azure AI Search, the framework is meant to let abstract layers support it. changing to different systems. Reimplementing the ingestion pipeline might use Google or AWS OpenSearch Service. Cloud-based Vertex AI Search with like properties. For companies trying to change, Omo offers a good open-source substitute. Supports self-hosted installations with connectors while providing an architecture free of LLM and vector-store dependencies. for Notion, Google Drive, OneDrive, and Confluence. Google Cloud Agent space also offers technology- Agnostic compatibility with more than 80 pre-built connections for outside programs including Jira, SharePoint, and Salesforce [12].  $\Phi$  Enduring Multidimensional Filtering State and AI-Verified Badge are the core cognitive UX strategies; critically speaking. Independent of the the basic search engine. Because of this abstraction, businesses could employ the framework independent of their current setup. framework of the cloud. Table 4 presents a comparative study of the Smart Search Portal against top businesses browse structures:

TABLE IV. COMPARATIVE FRAMEWORK ANALYSIS: SMART SEARCH PORTAL VS. COMPETING ENTERPRISE SEARCH SOLUTIONS

| Dimension            | Smart Search Portal (This Work) | Google Cloud Agent space                                 | Omo (Open-Source)            |
|----------------------|---------------------------------|----------------------------------------------------------|------------------------------|
| Cloud Platform       | Azure AI Search                 | Google Cloud (Vertex AI)                                 | Cloud-agnostic / Self-hosted |
| LLM Backend          | Azure OpenAI                    | Gemini                                                   | LLM-agnostic                 |
| Vector Store         | Azure Hybrid Index              | Google native                                            | Vector Store                 |
| Connectors           | SharePoint (Entra ID)           | 80+ pre-built (SharePoint, Salesforce, Jira, Confluence) | Connectors                   |
| CLT/IFT Design Layer | Yes (explicit)                  | No                                                       | CLT/IFT Design Layer         |

| Dimension             | Smart Search Portal (This Work)     | Google Cloud Agent space | Omo (Open-Source)     |
|-----------------------|-------------------------------------|--------------------------|-----------------------|
| Confidence Indicators | AI-Verified Badge ( $R \geq \tau$ ) | Not documented           | Confidence Indicators |
| Human-in-the-Loop     | Like/Dislike RLHF loop              | Not documented           | Human-in-the-Loop     |
| Open-Source           | No                                  | No                       | Open-Source           |

## VII. LIMITATIONS AND FUTURE WORK

### A. Limitations

Generalizability beyond engineering: The engineering industry tested this structure. In a semiconductive environment, paperwork. Its applicability in finance, medicine, and other sectors needs to be evaluated in future studies. Legal fields with varied cognitive and trust criteria apply. Long-term confidence calibration: The study examined the immediate results of trust over only one session. Tomorrow Research should find out if the AI-Verified Badge is still useful after a long time. especially when taking into account the directions of automation bias and algorithm aversion [14]. Performance measurements were developed utilising a 125,000-document index, and scalability beyond this level is expected. Enterprise-level testing (over 1 million documents) is necessary, especially in the indexing region. The request's throughput and latency for parallel workloads. The study was carried out by twenty engineers from the same firm. One should take cross-industry variations and demographic factors—age, digital literacy, and domain experience—into account. Researched in more varied, larger samples. validation via one platform. Though the architecture is intended for cloud-independent deployment, empirical validation was conducted. only done on Azure AI Search. Cross-platform validation is said to be the primary future study of emphasis. direction.

### B. Future Work

1. Longitudinal user studies to examine the growth of search behaviors and trust calibration over time. Long (three to six months) times of deployment.
2. Validate cross-platform with Google Cloud Vertex AI Search and AWS OpenSearch Service Check the general applicability of the cognitive UX solutions in a cloud-independent manner.
3. Fine-grained confidence calibration helps to distinguish between high- and low-risk situations to fit several corporate risk profiles [16]. draws technical details from conventional source documents.
4. Adaptive  $\Phi$  (Persistent Filtering State), which automatically fills filters based on how users behave. moving toward proactive rather than reactive personalization and historical behavior.
5. To let cooperative, multi-agent system integration as described by Cherukuri and Yarrar be possible. Research and verification procedures for dispersed corporate teams.
6. Open-source alternatives such Omo and Google Agent space versus empirical comparison

to give specialists evidence-based advice on platform selection.

## VIII. ETHICAL CONSIDERATIONS & FAIRNESS ASSURANCE

1) *Calibration of trust*: Through explicit confidence quantification, the AI-Verified Badge encourages appropriate trust. Training materials must guarantee that users comprehend:

- a) *The badge is probabilistic, not ground-truth;*
- b) *confidence scores may be wrong;*
- c) *important choices require human verification;*
- d) *system limits differ per document type.*

2) *Fairness Inspection Protocol: Prior to deployment:*

- a) *Analysis of demographic equity across several groups all sorts of papers.*
- b) *Evaluated probabilities (comparison of false positive/negative rates)*
- c) *Performing counterfactual testing to assess the embedding's sensitivity to features that are unrelated.*

3) *Monitoring of Production*: Automated alerts are included in monthly fairness reports if the disparity is greater than 15%.

- a) *Documents in each category are reviewed by experts 100 times every three months*
- b) *Conduct stratified analysis of like/dislike patterns across user demographics.*

4) *Mitigation*: Category-specific threshold ( $\tau$ ) calibration if bias detected; keep a public audit trail of all assessments and activities[15].

5) *Data Privacy*: According to the GDPR (Articles 5, 25, 32), user searches and comments are pseudonymized. Only operationally important information (Active Tickets, Role) was kept. Anonymization of data after 90 days. Users can choose to opt out without having an impact on functionality.

6) *Embedding Bias*: Regular WEAT/SEAT assessments for gender, race, and cultural biases; quarterly fairness re-evaluation; consider fine-tuning on diverse enterprise documents.

7) *Human-in-the-Loop Empowerment*: Feedback like "Like" or "Dislike" serves two ethical functions: 1) enabling users to actively participate in the system, and 2) identifying bias by analyzing the aggregate satisfaction patterns across demographics. This minimizes algorithmic antipathy while upholding a suitable level of doubt.

## IX. CONCLUSION

This paper addressed the growing challenge of cognitive overload in enterprise knowledge discovery from legacy, unmanaged document repositories. We presented a Hybrid Semantic Search Framework that integrates Cognitive Load Theory and Information Foraging Theory with a production-grade Azure AI Search implementation.

Key Contributions:

- a) *Empirical*: 91.4% reduction in Task Completion Time Two results: The Mental Demand score fell 36.1 points (Cohen's  $d=5.8$ ) ( $\eta^2=0.69$ ).

b) *Technical factors*: 87 ms P50 latency and 94.2% badge accuracy. One hundred twenty-five thousand papers recall 88.7 percent.

c) *Methodological*: five businesses undergo a methodical assessment, Repository Failure

This study offers a method that is both human-centered and scalable Enterprise knowledge management map, handles the primary design element should be cognitive UX rather than just a limitation on the recovery infrastructure. Upcoming initiatives will underline cross-platform authentication and longitudinal trust. Complete RLHF integration, calibration, and RLHF [3][14].

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